

Chapter 8

Aggregate Expenditure and Equilibrium Output

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Contents

- 1 The Core of Macroeconomic Theory
- 2 The Keynesian Theory
- 3 Equilibrium Output (Income)
- 4 The Multiplier



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Looking Ahead

CHAPTERS 8–9**The Goods-and-Services Market**

- Planned aggregate expenditure
Consumption (C)
Planned investment (I)
Government (G)
- Aggregate output (income) (Y)

CHAPTER 11**Full Equilibrium: AS/AD Model**

- Aggregate supply curve
- Fed rule
- Aggregate demand curve

Equilibrium interest rate (r^*)
Equilibrium output (income) (Y^*)
Equilibrium price level (P^*)

CHAPTER 13**The Labor Market**

- The supply of labor
- The demand for labor
- Employment and unemployment

CHAPTERS 10**The Money Market**

- The supply of money
- The demand for money
- Interest rate (r)

CHAPTER 12**Policy and Cost Effects in the AS/AD model**

▲ **FIGURE V.1 The Core of Macroeconomic Theory**

We build up the macroeconomy slowly. In Chapters 8 and 9, we examine the market for goods and services. In Chapter 10, we examine the money market. Chapter 11 introduces the aggregate supply (AS) curve and the Fed rule and derives the aggregate demand (AD) curve. Chapter 12 uses the AS/AD model to examine policy and cost effects, and Chapter 13 discusses the labor market.

The Keynesian Theory

- Classical economics, which was aggregate-supply-focused, could not explain why such high unemployment prolonged during 1930s.
- Keynes published *The General Theory of Employment, Interest, and Money* in 1936, and argued the important role of **aggregate demand** in determining aggregate income in the economy.

$$\underbrace{AE}_{\text{Planned Aggregate Expenditure}} = C + I + G + \underbrace{EX - IM}_{\text{Net Export}}.$$

- Model Setting
 - Assumption: price level (P) and interest rates are fixed.
 - Aggregate output (Y) is in real term.

Consumption C

- Consumption function represents the relationship between consumption and income.

People are disposed to increase their consumption as their incomes increase, but not by as much as the increase in their income.

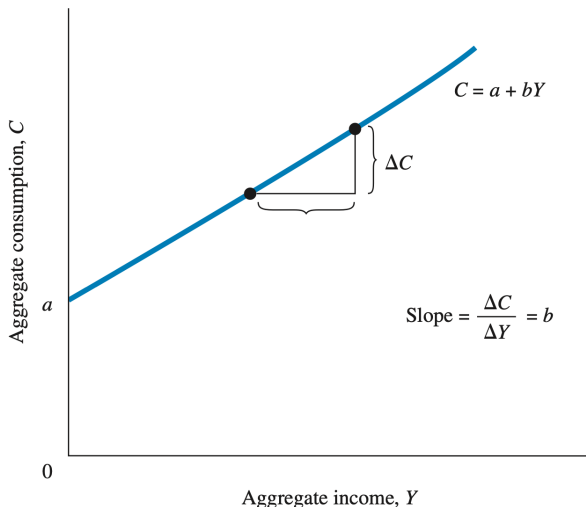
Consumption C

- Consumption function represents the relationship between consumption and income.

People are disposed to increase their consumption as their incomes increase, but not by as much as the increase in their income.

$$\begin{aligned}C &= C(Y) \\ &= a + bY \\ \text{where } a &> 0, \ b < 1.\end{aligned}$$

Keynesian Consumption Function



MPC and MPS

- Marginal propensity to consume

$$\text{MPC} \equiv \text{slope of consumption function} \equiv \frac{\Delta C}{\Delta Y}.$$

- Aggregate savings (S)

$$S \equiv Y - C.$$

- Marginal propensity to save

$$\text{MPS} \equiv 1 - \text{MPC}.$$

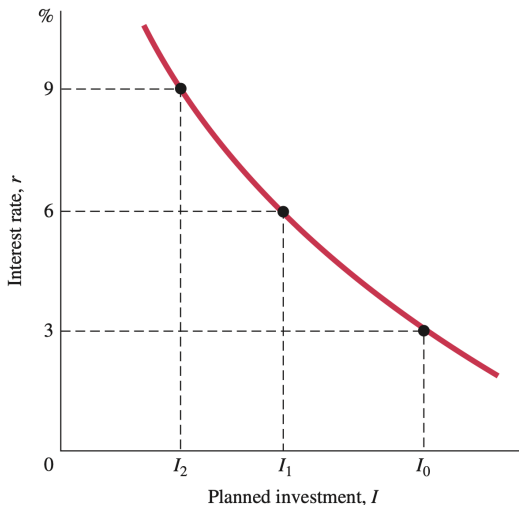
Other Determinants of Consumption

- The assumption that consumption depends only on income is a simplification.
- The decisions of households on how much to consume are also affected by
 - wealth
 - interest rate
 - expectations of the future

Investment I

- Investment: addition to capital stock of the economy
 - Capital: factories, machines, or any item that is used to produce other goods and services.
- Planned investment (I) versus actual investment (I^a)
- Planned investment and the interest rate (r)
 - Firms' decision to invest depends on whether the expected profits can justify its costs.
 - One of the costs of an investment project is the interest rate.
- Other determinants of planned investment
 - "*Animal spirits*": firms' expectation of future sales

Planned Investment Schedule



The Determination of Equilibrium Output (Income)

- The economy is defined to be in equilibrium when aggregate output (Y) is equal to planned aggregate expenditure (AE).

$$\begin{aligned}\text{Equilibrium : } Y &= AE \\ &= C + I.\end{aligned}$$

- What happens if the economy is out of equilibrium

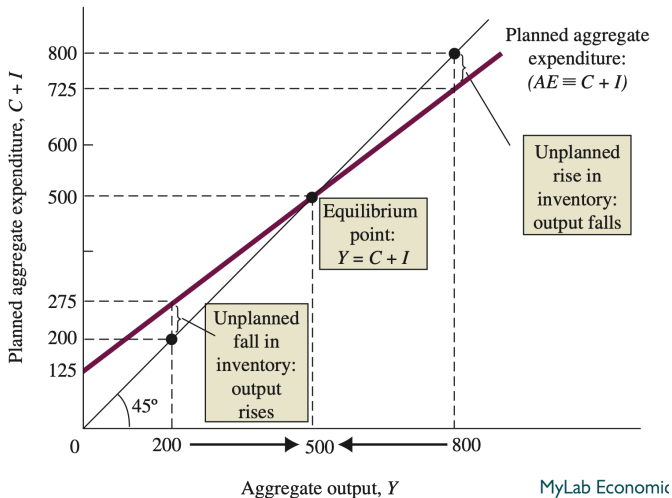
$$Y > C + I$$

→ unplanned inventory investment increases.

$$Y < C + I$$

→ unplanned inventory investment decreases.

Graphical Derivation of Equilibrium Output



Algebraic Derivation of Equilibrium Output

One unknown, one equation

$$\begin{aligned} Y &= AE \\ &= C + I \\ &= a + bY + I. \end{aligned}$$

$$\begin{aligned} (1 - b)Y &= a + I \\ \rightarrow Y^* &= \frac{a + I}{1 - b}. \end{aligned}$$

The Saving/Investment Approach to Equilibrium

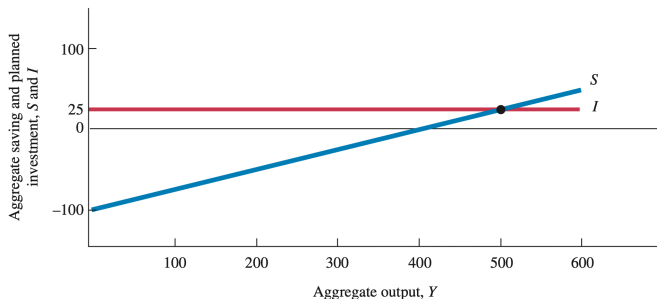
- Aggregate income must be saved or spent.

$$Y \equiv C + S, \text{ by definition.}$$

- By substituting $C + S$ for Y in the equilibrium condition,

$$C + S = C + I.$$

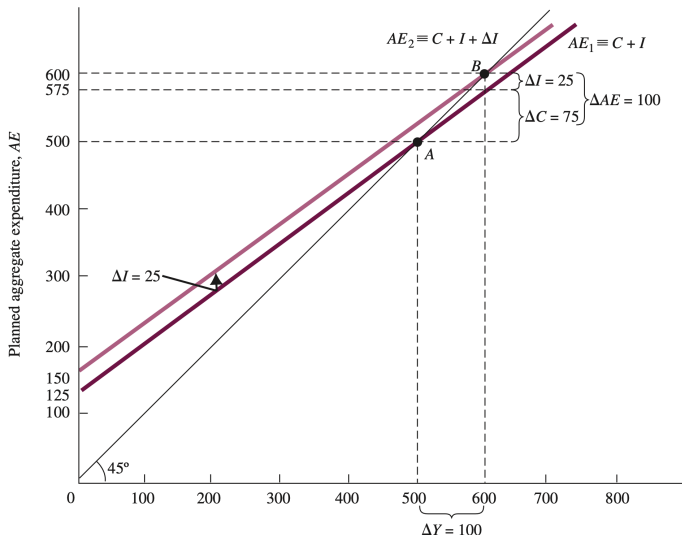
$$\rightarrow S = I.$$



An Effect of Changes in I on Y^*

- Multiplier: the ratio of the change in the equilibrium level of output to a change in some exogenous variable.
 - Exogenous variable: a variable that is assumed not to depend on the state of the economy—that is, it does not change when the economy changes.
- Consider a sustained increase in planned investment of 25 ($I = 25, I' = 50$).

Graphic Analysis



The Multiplier Equation

- Recall that the marginal propensity to save (MPS) is the fraction of a change in income that is saved

$$MPS = \frac{\Delta S}{\Delta Y}.$$

- In equilibrium, S has to be equal to I .

$$MPS = \frac{\Delta I}{\Delta Y}.$$

- Therefore,

$$\Delta Y = \frac{\Delta I}{MPS}.$$

- Finally, because $MPS + MPC \equiv 1$,

$$\Delta Y = \frac{\Delta I}{1 - MPC}.$$